SOLUTIONS FOR WORKSHEET 3

Question 1. Using the algebraic limit theorem, write down the limits of (a_n):

(a) $a_n = \frac{1}{n} + \frac{6}{n^3}$ (b) $a_n = \frac{n^2-5}{n^2+6n}$ (c) $a_n = \frac{5n^2+3n-2}{2n^2+7n+3}$

(d) $a_n = \frac{-2n^3+4n^2+1}{7n^3+6n^2+5n+4}$ (e) $a_n = \frac{n^5+n^3+n}{n^7+n^5+n^3+n}$

Solution. (a) $\lim\left(\frac{1}{n} + \frac{6}{n^3}\right) = \lim\left(\frac{1}{n}\right) + 6\lim\left(\frac{1}{n^3}\right) = 0 + 6(0) = 0$

(b) $\lim\left(\frac{n^2-5}{n^2+6n}\right) = \lim\left(\frac{1-\frac{5}{n^2}}{1+\frac{6}{n}}\right) = \frac{1-0}{1+0} = 1$ (c) $\lim\left(\frac{5n^2+3n-2}{2n^2+7n+3}\right) = \lim\left(\frac{5+\frac{3}{n}-\frac{2}{n^2}}{2+\frac{7}{n}+\frac{3}{n^2}}\right) = \frac{5+0-0}{2+0+0} = \frac{5}{2}$

(d) $\lim\left(\frac{-2n^3+4n^2+1}{7n^3+6n^2+5n+4}\right) = \lim\left(\frac{-2+\frac{4}{n}+\frac{1}{n^3}}{7+\frac{6}{n}+\frac{5}{n^2}+\frac{4}{n^3}}\right) = -\frac{2}{7}$ (e) $\lim\left(\frac{n^5+n^3+n}{n^7+n^5+n^3+n}\right) = \lim\left(\frac{\frac{1}{n^2}+\frac{1}{n^4}+\frac{1}{n^6}}{1+\frac{1}{n^2}+\frac{1}{n^4}+\frac{1}{n^6}}\right) = 0$

Question 2. Give an example of a convergent sequence that is not monotone. Additionally, give an example of a monotone sequence that is not convergent.

Answer. $\left(\frac{(-1)^n}{n}\right)$ is convergent but not monotone. (n) is monotone but diverges.

Question 3. When does a monotone increasing sequence converge? When does a monotone decreasing sequence converge?

Answer. (MCT) When a monotone increasing/decreasing sequence is bounded above/below, then it is convergent.

Question 4. Let (a_n) and (b_n) be divergent sequences. Give example of sequences such that the following limits exist. Otherwise, say the limit cannot exist.

(a) $\lim(ca_n)$ for all $c \in \mathbb{R}$ (b) $\lim(a_n + b_n)$

(c) $\lim(a_n b_n)$ (d) $\lim\left(\frac{a_n}{b_n}\right)$ with $b_n \neq 0$ for all $n \in \mathbb{N}$

Answer. (a) If $c = 0$ then $(ca_n) = (0) \Rightarrow \lim(ca_n) = 0$.

(b) If $(a_n) = ((-1)^{n+1})$ and $(b_n) = ((-1)^n)$ then both (a_n) and (b_n) diverge but $(a_n + b_n) = (0)$ and this limit is 0.

(c) If $(a_n) = (b_n) = ((-1)^n)$, then both (a_n) and (b_n) diverge but $(a_n b_n) = (1)$ has limit 1.

(d) If $(a_n) = (b_n) = ((-1)^n)$, then then both (a_n) and (b_n) diverge but $\left(\frac{a_n}{b_n}\right) = (1,1,1,\ldots)$ has limit 1.

Question 5. Let (a_n) and (b_n) be sequences, both diverge to positive infinity. Prove or disprove:

(a) $\lim(ca_n) = \begin{cases} \infty, & c \in \mathbb{R}^+ \\ -\infty, & c \in \mathbb{R}^- \end{cases}$ (b) $\lim(a_n + b_n) = \infty$ (c) $\lim(a_n b_n) = \infty$.

(d) $\lim\left(\frac{a_n}{b_n}\right) = k =$ a fixed number, provided $b_n \neq 0\ \forall n \in \mathbb{N}$.

Proof. *We introduce notations* $\forall$ *(for all),* $\exists$ *(there exist(s)) and* $\ni$ *(such that) to provide an alternative in the writing of the proofs.*

(a) Case 1. $c > 0$. Given that $\forall M > 0, \exists N \in \mathbb{N} \ni a_n > \frac{1}{c}M\ \forall n > N$. Then

$$\forall M > 0, \exists N \in \mathbb{N} \ni ca_n > M\ \forall n > N.$$

Case 2. $c < 0$. Let $M < 0$. Given that $\forall \frac{1}{c}M > 0, \exists N \in \mathbb{N} \ni a_n > \frac{1}{c}M \forall n > N$. Then

$$\forall M < 0, \exists N \in \mathbb{N} \ni ca_n < M\ \forall n > N.$$

(b) Given that $\begin{cases} \forall M > 0, \exists N_1 \in \mathbb{N} \ni a_n > \frac{1}{2}M \forall n > N_1 \\ \forall M > 0, \exists N_2 \in \mathbb{N} \ni b_n > \frac{1}{2}M \forall n > N_2 \end{cases}$. Then

$$\forall M > 0, \exists N = \max\{N_1, N_2\} \in \mathbb{N} \ni a_n + b_n > M\ \forall n > N.$$

(c) Given that $\begin{cases} \forall M > 0, \exists N_1 \in \mathbb{N} \ni a_n > \sqrt{M} \forall n > N_1 \\ \forall M > 0, \exists N_2 \in \mathbb{N} \ni b_n > \sqrt{M} \forall n > N_2 \end{cases}$. Then

$$\forall M > 0, \exists N = \max\{N_1, N_2\} \in \mathbb{N} \ni a_n b_n > M\ \forall n > N.$$

(d) False: $(a_n) = (n) = (b_n) \Rightarrow \lim\left(\frac{a_n}{b_n}\right) = 1$; but $(\tilde{a}_n) = (n), (\tilde{b}_n) = (n^2) \Rightarrow \lim\left(\frac{\tilde{a}_n}{\tilde{b}_n}\right) = 0$.

Question 6. Use the squeeze theorem to find the following limits.

(a) $\lim\left(\dfrac{\sin 2n}{n}\right)$ (b) $\lim\left(\dfrac{\sin 2n}{n^2 \cos n}\right)$ (c) $\lim\left(\dfrac{\ln n}{n}\right)$ (Hint: $0 \leq \ln n \leq \sqrt{n-1}\ \forall n \in \mathbb{N}$)

Solution. The solutions are true for all $n \in \mathbb{N}$:

(a) $-1 \leq \sin 2n \leq 1 \Rightarrow \dfrac{-1}{n} \leq \dfrac{\sin 2n}{n} \leq \dfrac{1}{n} \Rightarrow \lim \dfrac{\sin 2n}{n} = 0$ as $\lim\left(\dfrac{-1}{n}\right) = 0 = \lim\left(\dfrac{1}{n}\right)$.

(b) $-1 \leq \sin n \leq 1 \Rightarrow \dfrac{-1}{n^2} \leq \dfrac{\sin n}{n^2} \leq \dfrac{1}{n^2} \Rightarrow \lim \dfrac{\sin n}{n^2} = 0$ as $\lim\left(\dfrac{-1}{n^2}\right) = 0 = \lim\left(\dfrac{1}{n^2}\right)$.

Now, $\lim \dfrac{\sin 2n}{n^2 \cos n} = \lim\left(\dfrac{2 \cos n \sin n}{n^2 \cos n}\right) = \lim\left(\dfrac{2 \sin n}{n^2}\right) = 0$.

(c) $0 \leq \ln n \leq \sqrt{n-1} \Rightarrow 0 \leq \dfrac{\ln n}{n} \leq \dfrac{\sqrt{n-1}}{n} = \sqrt{\dfrac{1}{n} - \dfrac{1}{n^2}} \Rightarrow \lim\left(\dfrac{\ln n}{n}\right) = 0$ because

$\lim\left(0\right) = 0 = \lim\left(\sqrt{\frac{1}{n} - \frac{1}{n^2}}\right)$.

Question 7. Use Corollary 8.4 to show that the sequences below are divergent.

(a) $\left(1+(-1)^n\right)$ (b) $\left(\cos n\pi\right)$ (c) (1, 1, 1, 2, 1, 3, 1, 4, 1, 5, 1, 6, …)

Solution. (a) Choose the following two subsequences

$$\left(a_{n_k}\right)=(0,\ 0,\ 0,\ \ldots\ )\to 0 \text{ where } \left(n_k\right)=\left(1,3,5,\ldots\right)$$

$$\text{and } \left(a_{\tilde{n}_k}\right)=(2,\ 2,\ 2,\ \ldots)\to 2 \text{ where } \left(\tilde{n}_k\right)=\left(2,4,6,\ldots\right).$$

Both subsequences converge differently. Hence, by Corollary 8.4, $\left(1+(-1)^n\right)$ diverges.

(b) Choose the following two subsequences

$$\left(a_{n_k}\right)=(-1,\ -1,\ -1,\ \ldots\ )\to -1 \text{ where } \left(n_k\right)=\left(1,3,5,\ldots\right)$$

$$\text{and } \left(a_{\tilde{n}_k}\right)=(1,\ 1,\ 1,\ \ldots)\to 1 \text{ where } \left(\tilde{n}_k\right)=\left(2,4,6,\ldots\right).$$

Both subsequences converge differently. Hence, by Corollary 8.4, $\left(\cos n\pi\right)$ diverges.

(c) (1, 1, 1, 2, 1, 3, 1, 4, 1, 5, 1, 6, …)

Choose the subsequence (1, 2, 3, 4, 5, 6, …) $\to\infty$ where $\left(\tilde{n}_k\right)=\left(2,4,6,\ldots\right)$. Hence, by (contrapositive of) Theorem 8.3, the parent sequence diverges.

Question 8. Prove Theorem 6.3 part (ii) and (iii) from the notes.

Let (a_n) and (b_n) be convergent sequences with $\lim(a_n)=a$ and $\lim(b_n)=b$. Then,

(i) $a_n\geq 0\ \forall n\in\mathbb{N}\Rightarrow a\geq 0$. (ii) $a_n\leq b_n\forall n\in\mathbb{N}\Rightarrow a\leq b$.

(iii) $\exists c\in\mathbb{R}\ni c\leq b_n\forall n\in\mathbb{N}\Rightarrow c\leq b$. (iv) If $\exists c\in\mathbb{R}\ni a_n\leq c\ \forall n\in\mathbb{N}\Rightarrow a\leq c$.

Proof. (ii) Let $c_n=b_n-a_n$. Then from (i) and the fact that $c_n\geq 0\ \forall n\in\mathbb{N}$, it then follows that $c=b-a\geq 0$.

(iii) Let $(c_n)=(c)$. If $\exists c\in\mathbb{R}\ni c=c_n\leq b_n\forall n\in\mathbb{N}$, then by (ii), $b\geq c$.

(iv) Let $(c_n)=(c)$. If $\exists c\in\mathbb{R}\ni c=c_n\geq a_n\forall n\in\mathbb{N}$, then by (ii), $c\geq a$.

Additional explanation for Question 8

Let $\left(s_n\right),\left(a_n\right),\left(b_n\right),\left(c_n\right)$ be convergent sequences that converge to s, a, b, and c, respectively, i.e., $\lim\left(s_n\right)=s,\lim\left(a_n\right)=a,\lim\left(b_n\right)=b$, and $\lim\left(c_n\right)=c$. Prove that

(i) $s_n\geq 0$ for all $n\in\mathbb{N}$ implies that $s\geq 0$.

(ii) $a_n\geq b_n$ for all $n\in\mathbb{N}$ implies that $a\geq b$.

(iii) there exists $k\in\mathbb{R}$ such that $a_n\geq k$ for all $n\in\mathbb{N}$ implies that $a\geq k$.

(iv) there exists $k\in\mathbb{R}$ such that $b_n\leq k$ for all $n\in\mathbb{N}$ implies that $b\leq k$.

Proof. (i) Assume that $s<0$. Then by the definition of convergence of sequence, for all $\varepsilon>0$, there exists an $N\in\mathbb{N}$ such that $n>N\Rightarrow\left|s_n-s\right|<\varepsilon$. Pick $\varepsilon=\left|s\right|=-s$. Then $\left|s_n-s\right|<-s\Rightarrow s+s<s_n<s-s\Rightarrow 2s<s_n<0$, a contradiction. Hence $s\geq 0$.

(ii) Let $s_n=a_n-b_n\ \forall n\in\mathbb{N}$. Then $s_n\geq 0$ because $a_n\geq b_n\ \forall n\in\mathbb{N}$. By (i), $s\geq 0$. By Theorem 6.1, $s=a-b$. Hence, $a\geq b$.

(iii) Set $\left(c_n\right):=\left(k\right)$ and let $s_n=a_n-c_n\ \forall n\in\mathbb{N}$. Then $s_n\geq 0$ since $a_n\geq k\ \forall n\in\mathbb{N}$. By (i), $s\geq 0$. By Theorem 6.1, $s=a-c=a-k$. Hence, $a\geq k$.

(iv) Set $\left(c_n\right):=\left(k\right)$ and let $s_n=c_n-b_n\ \forall n\in\mathbb{N}$. Then $s_n\geq 0$ since $b_n\leq k\ \forall n\in\mathbb{N}$. By (i), $s\geq 0$. By Theorem 6.1, $s=b-c=b-k$. Hence, $b\leq k$.

Remark: 1. $\left(c_n\right):=\left(k\right)$ is a constant sequence, so $\lim\left(c_n\right)=c=k$.

2. When we let $s_n=a_n-b_n$, or $s_n=a_n-c_n$, or $s_n=c_n-b_n$; these are new sequences $\left(a_1-b_1,a_2-b_2,a_3-b_3,\ldots\right),\left(a_1-c_1,a_2-c_2,a_3-c_3,\ldots\right),\left(c_1-b_1,c_2-b_2,c_3-b_3,\ldots\right)$.

3. Let $\alpha,\beta\in\mathbb{R}$ be fixed constants. Theorem 6.1 states that if $\left(a_n\right),\left(b_n\right)$ converge to a and b, respectively, then $\lim\left(\alpha a_n\pm\beta b_n\right)=\alpha a\pm\beta b$, $\lim\left(a_n b_n\right)=ab$, and $\lim\left(\frac{a_n}{b_n}\right)=\frac{a}{b}$, where $b_n>0\ \forall n\in\mathbb{N}$.

4. It is also true that $\lim\left(\left(a_n\right)^k\right)=a^k$, $\lim\left(\sqrt[k]{a_n}\right)=\sqrt[k]{a}$, $\lim\left(k^{a_n}\right)=k^a$, where $k>0$ is a fixed real number.

Question 9. Prove the if ($\Leftarrow$) statement in the Theorem 8.3.

Theorem 8.3. A sequence (a_n) converges if and only if every subsequence of (a_n) converges (to $\lim(a_n)$).

Proof. $\Leftarrow$) If every subsequence of (a_n) converges (to $\lim(a_n)$), then take the subsequence equals to the parent sequence, i.e., $\left(a_{n_k}\right)=\left(a_n\right)$, and so (a_n) converges (to $\lim(a_n)$).

Question 10. Let (b_n) be a monotone increasing sequence with $b_1 \geq 0$. Define a new sequence (a_n) by $a_n = \frac{2b_n+3}{b_n+2}$. (a) Show that (a_n) is monotone increasing; (b) Prove that (a_n) is bounded and hence show it is convergent; (c) Find $\lim(a_n)$.

Proof. Note that $b_n \geq 0 \ \forall n \in \mathbb{N}$. Write $a_n = 2 - \frac{1}{b_n+2}$.

(a) $a_{n+1} - a_n = \frac{1}{b_n+2} - \frac{1}{b_{n+1}+2} = \frac{b_{n+1}-b_n}{(b_{n+1}+2)(b_n+2)} \geq 0$ as (b_n) is increasing. So, (a_n) is monotonic increasing.

(b) $0 < 2 - a_n = \frac{1}{b_n+2} \leq \frac{1}{2} \ \forall n \in \mathbb{N}$. So, (a_n) is bounded. By MCT, (a_n) converges.

(c) If (b_n) is bounded then $\lim(b_n) = b$ and $\lim(a_n) = 2 - \frac{1}{b+2}$. Otherwise, $\lim(b_n) = \infty$ and then $\lim(a_n) = 2$.

Question 11. State the Bolzano-Weierstrass theorem.

(a) Give an example of a sequence (a_n), where $6 \leq a_n \leq 8 \ \forall n \in \mathbb{N}$ and (a_n) is divergent. Using this example, write down a subsequence of (a_n) that converges.

(b) Let (b_n) be the sequence given by $b_n = \frac{3n\cos(2n^2+1)}{n+2}$. Explain why (b_n) has a convergent subsequence.

Solution. BWT: Every bounded sequence has a convergent subsequence.

(a) $(a_n) = (7 + (-1)^n)$, then (a_n) is divergent (and bounded). Hence, we can form at least one subsequence which is convergent, for example (6, 6, 6, ...) or (8, 8, 8, ...).

(b) $-3 < -3 + \frac{6}{n+2} = -\frac{3n}{n+2} \leq \frac{3n\cos(2n^2+1)}{n+2} \leq \frac{3n+6-6}{n+2} = 3 - \frac{6}{n+2} < 3$. So, (b_n) is bounded and by BWT, it has a convergent subsequence.

Question 12. Given an example of a sequence (a_n) with:

- a subsequence that converges to 1,
- another subsequence that converges to 17,
- and a final subsequence that converges to π.

Answer. (1, 17, π, 1, 17, π, ...)

Question 13. Prove that the sequences converge and find their limit.

(a) $(a_n)=\left(\sqrt{2},\sqrt{2+\sqrt{2}},\sqrt{2+\sqrt{2+\sqrt{2}}},...\right)$ (b) $(b_n)=\left(\sqrt{2},\sqrt{2\sqrt{2}},\sqrt{2\sqrt{2\sqrt{2}}},...\right)$

Solution. (a) Note that $a_n>0\ \forall n\in\mathbb{N}$. Write

$$a_{n+1}=\sqrt{2+a_n}\ \forall n\in\mathbb{N}\ \text{ with }\ a_1=\sqrt{2}\ \text{ and }\ a_2=\sqrt{2+\sqrt{2}}\,.$$

First, we prove that $a_n\geq\sqrt{2}\ \ \forall n\in\mathbb{N}$. It is clear that $a_1=\sqrt{2}$. Now assume that $a_k>\sqrt{2}$. Then $a_{k+1}^2=2+a_k>2+\sqrt{2}>2\Rightarrow a_{k+1}>\sqrt{2}$. Hence, $a_n\geq\sqrt{2}\ \ \forall n\in\mathbb{N}$.

Next, we prove that $a_n<2\ \ \forall n\in\mathbb{N}$. It is clear that $a_1=\sqrt{2}<2$. Now assume that $a_k<2$. Then $a_{k+1}^2=2+a_k<2+2=4\Rightarrow a_{k+1}<2$. Hence, $\sqrt{2}\leq a_n<2\ \ \forall n\in\mathbb{N}$.

So, the sequence is bounded. Now

$$a_{n+1}-a_n=\sqrt{2+a_n}-\sqrt{2+a_{n-1}}=\frac{a_n-a_{n-1}}{\sqrt{2+a_n}+\sqrt{2+a_{n-1}}}\,.$$

By induction, if $a_n-a_{n-1}>0$, then $a_{n+1}-a_n>0$. Now since $a_2-a_1>0$. So we conclude that the sequence is increasing. Now, by MCT, the sequence converges. Let $\lim(a_n)=a\Rightarrow\lim(a_{n+1})=a$. Solving $a^2=2+a$ we obtain $a=2\ \ (a>0)$.

(b) Note that $b_n>0\ \forall n\in\mathbb{N}$. Write $b_{n+1}=\sqrt{2b_n}\ \ \forall n\in\mathbb{N}$ with $b_1=\sqrt{2}$ and $b_2=\sqrt{2\sqrt{2}}$.

First, we prove that $b_n\geq\sqrt{2}\ \ \forall n\in\mathbb{N}$. It is clear that $b_1=\sqrt{2}$. Now assume that $b_k>\sqrt{2}$. Then $b_{k+1}^2=2b_k>2\sqrt{2}>2\Rightarrow b_{k+1}>\sqrt{2}$. Hence, $b_n\geq\sqrt{2}\ \ \forall n\in\mathbb{N}$.

Next, we prove that $b_n<2\ \ \forall n\in\mathbb{N}$. It is clear that $b_1=\sqrt{2}<2$. Now assume that $b_k<2$. Then $b_{k+1}^2=2b_k<2\times2=4\Rightarrow b_{k+1}<2$. Hence, $\sqrt{2}\leq b_n<2\ \ \forall n\in\mathbb{N}$.

So, the sequence is bounded.

Now $$b_{n+1}-b_n=\sqrt{2b_n}-\sqrt{2b_{n-1}}=\frac{2(b_n-b_{n-1})}{\sqrt{2b_n}+\sqrt{2b_{n-1}}}\,.$$

By induction, if $b_n-b_{n-1}>0$, then $b_{n+1}-b_n>0$. Now since $b_2-b_1>0$. So we conclude that the sequence is increasing. Now, by MCT, the sequence converges. Let $\lim(b_n)=b\Rightarrow\lim(b_{n+1})=b$. Solving $b^2=2b$ we obtain $b=2\ \ (a>0)$.

Question 14. Prove that the sequence $(a_n)=(x^n)$ converges if $-1<x\le 1$, where $\lim(a_n)=\begin{cases}0, & |x|<1\\ 1, & x=1\end{cases}$; and the sequence diverges if otherwise.

Proof. Case 1. If $x=1$ or $x=0$, then $(a_n)=(1)$ or $(a_n)=(0)$ are convergent (constant) sequences which converge to 1 or 0.

Case 2. When $0<x<1$, then for all $n\in\mathbb{N}$,

$$x^n-x^{n+1}=x^n\left(1-x\right)>0 \text{ because } 0<x<1 \text{ and } x^n>0.$$

Now because $x>x^2>x^3>\cdots>x^n>\cdots>0$, the sequence $(a_n)=(x^n)$ is decreasing and bounded below. Hence, by MCT, the sequence converges to $\lim(x^n)=a$. To compute a, note that (x^{2n}) is a subsequence so by Theorem 8.3, $\lim(x^{2n})=a$. Now, by Theorem 6.1 (c), and Theorem 4.9 (uniqueness of limit) and the sequence is strictly decreasing,

$$a=\lim(x^{2n})=\lim(x^n)\times\lim(x^n)=a^2\Rightarrow a=0.$$

Case 3. For $-1<x<0$, we claim that $\lim(x^n)=0$. We want to show this (by using the definition): $\forall\varepsilon>0,\exists N\in\mathbb{N}\ni|x^n-0|<\varepsilon$ for all $n>N$ (where $-1<x<0$). But this is equivalent to $|x^n-0|<\varepsilon$ for $0<x<1$. Hence, by Case 2, $\lim(x^n)=0$.

Case 4. For $x>1$ or $x<-1$, $(a_n)=(x^n)$ is unbounded and therefore divergent.

Case 5. For $x=-1$, $(a_n)=((-1)^n)$ is divergent.

Question 15. Let (a_n) and (b_n) be Cauchy and let $c_n=|a_n-b_n|\forall n\in\mathbb{N}$. Prove that (c_n) is Cauchy.

Proof. $\begin{cases}\forall\varepsilon>0,\exists M\in\mathbb{N}\ni|a_n-a_m|<\frac{\varepsilon}{2}\forall m,n>M\\ \forall\varepsilon>0,\exists N\in\mathbb{N}\ni|b_n-b_m|<\frac{\varepsilon}{2}\forall m,n>N\end{cases}$. Therefore, for $m,n>\max\{N,M\}$, it implies that $|c_n-c_m|=|(a_n-a_m)-(b_n-b_m)|\le|a_n-a_m|+|b_n-b_m|<\varepsilon$.

Question 16.

(a) Using the monotone convergence theorem, prove without the use of the completeness axiom the Archimedean property of the real numbers.

(b) Using the monotone convergence theorem, prove without the use of the completeness axiom the nested interval property.

Proof. (a) We show that $\mathbb{N}$ are unbounded in $\mathbb{R}$. Assume the contrary. First, we note that $(a_n) = (n)$ forms an increasing sequence. If (a_n) is bounded, then by MCT, (a_n) converges (to $\lim(a_n) = a$). Now consider a subsequence $\left(a_{n_k}\right) = (n+1)$ (which means $\left(n_k\right) = (2,3,4,...)$). Then $\lim(n+1) = a+1$ by Theorem 6.1 but this contradicts to Theorem 8.3. Hence, $\mathbb{N}$ are unbounded in $\mathbb{R}$.

(b) Let $I_n = \left[a_n, b_n\right]$ such that $I_{n+1} \subset I_n \forall n \in \mathbb{N}$. Then (a_n) is a bounded and increasing sequence and (b_n) is a bounded and decreasing sequence because

$$a_1 \le a_2 \le \cdots \le a_n \le b_n \le \cdots \le b_2 \le b_1 .$$

Hence, (a_n) and (b_n) are convergent sequences. Let $\lim(a_n) = a$ and $\lim(b_n) = b$. Then by Theorem 6.3 (ii), $a \le b$. Hence, $\phi \ne \left[a, b\right] \subset \bigcap_{n=1}^{\infty} I_n$.

Question 17. Prove that every nonempty subset of the real numbers that is bounded above has a supremum directly from the Bolzano-Weierstrass theorem.

Proof. We use the following two lemmas to prove the theorem.

Lemma 1. The Bolzano–Weierstrass Theorem implies the Nested Interval Property.

Proof. Let $I_n = \left[a_n, b_n\right]$ such that $I_{n+1} \subset I_n \forall n \in \mathbb{N}$. For each n,

$$a_1 \le a_2 \le \cdots \le a_n \le b_n \le \cdots \le b_2 \le b_1 ,$$

so the sequence (b_n) is bounded. By the Bolzano–Weierstrass Theorem, it contains a convergent subsequence $\left(b_{n_k}\right) \to b$. We claim that

$$b \le b_n \forall n \in \mathbb{N} .$$

To show this, note that (b_n) and $\left(b_{n_k}\right)$ are both decreasing sequences since the intervals are nested.

If $b > b_m$ for some m, then for all $k \ge m$, we have $n_k \ge n_m \ge m$, and so

$$b_{n_k} \le b_{n_m} \le b_m < b \text{ and } \left|b_{n_k} - b\right| = b - b_{n_k} \ge b - b_m .$$

This is clearly impossible since $b - b_m$ is a fixed positive number and the sequence $\left(b_{n_k}\right)$ converges to b. Therefore, $b \le b_n \forall n \in \mathbb{N}$.

On the other hand, $b \geq a_n \forall n \in \mathbb{N}$. This is because, for any n, $b_{n_k} \geq a_n$, so, by Theorem 6.3(ii), $b = \lim\left(a_{n_k}\right) \geq a_n$. Therefore, $a_n \leq b \leq b_n \quad \forall n \in \mathbb{N}$.

Thus, $b \in I_n \forall n \in \mathbb{N}$. This means that $b \in \bigcap_{n=1}^{\infty} I_n$, so the intersection is non-empty.

Since our choice of nested intervals was arbitrary, we conclude that the Nested Interval Property is true.

Lemma 2. The Nested Interval Property implies the Axiom of Completeness.

Let A be a bounded set. We want to search for sup A and then use NIP to prove the supremum exists.

Proof. Let U_1 and L_1 be an upper bound and lower bound of A, respectively. Set $I_1 = [L_1, U_1]$ be the starting "area" of our search. The search gives $I_{n+1} \subset I_n \forall n \in \mathbb{N}$ with length proportionally $\left(\frac{1}{2}\right)^n$.

Applying NIP, $\bigcap_{n=1}^{\infty} I_n \neq \phi$. As the length $\left(\frac{1}{2}\right)^n \to 0$, there is a single $s \in \bigcap_{n=1}^{\infty} I_n$ which must be the least upper bound since $I_n = \left[L_n, U_n\right]$ which then implies that $L_n \leq s \leq U_n \ \forall n \in \mathbb{N}$. Thus, $s = \sup A$ because $s \geq L_n$ implies that s is an upper bound; and $s \leq U_n$ implies that s is the least upper bound.